

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 – RADIOACTIVITY AND STABILITY OF ISOTOPES (3.1): FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Use everything you have learned from Lesson 1 to Lesson 7 to explain the case of 'The Invisible Killer in the Rock': a piece of pitchblende (uranium ore) placed next to photographic film in complete darkness left a clear exposure on the film. Answer each section in the space under its prompt. Write in full sentences, use correct science words (nucleus, isotope, belt of stability, half-life, alpha, beta, gamma), and write nuclear equations where asked. Always link your ideas back to the rock. Use Kenyan examples from our lessons where you can. Your teacher will mark your work using the rubric.

Part 1: Why is the rock radioactive?

Our driving question asks: 'Why do some atoms in Kenyan rocks spontaneously release invisible energy?' (a) Explain what the nucleus of an atom is made of and what the terms atomic number (Z), mass number (A) and neutron number (N) mean. (b) Explain what an isotope is, using Carbon-12 and Carbon-14 as your example. (c) Using the idea of the belt of stability on an N vs Z graph, explain why the uranium-238 in the pitchblende is different from stable nuclei, and why this is a nuclear process and not a chemical one.

Write your answer here:

Part 2: What is coming out of the rock?

(a) Name the three types of radiation that unstable nuclei can emit. For each, state its charge, its relative penetrating power and what stops it. (b) Alpha radiation has the highest ionising ability, yet it is considered the least penetrating. Explain how both can be true. (c) Explain why alpha particles from the ore are still dangerous in a Kenyan mining situation even though paper stops them.

Write your answer here:

Part 3: Nuclear equations and the stability plot

(a) Write the balanced nuclear equation for the alpha decay of uranium-238 into thorium. (b) Write the nuclear equation for the beta-minus decay of thorium-234 into protactinium. (c) State what happens to A and Z in alpha decay, beta-minus decay and gamma emission. (d) Explain, using the N vs Z graph, how the decays of uranium-238 and its daughters move the nucleus on the plot, and what this shows about the nucleus and the belt of stability.

Write your answer here:

Part 4: Detecting the radiation and how long it lasts

(a) Explain how a Geiger-Müller counter turns invisible radiation from the rock into a click. (b) A GM counter clicks even when no source is nearby. Explain why, and how you should handle this when measuring the rock. (c) Define half-life. Explain, using your dice simulation and the half-life of uranium-238, why the pitchblende rock has been emitting radiation for billions of years. (d) A sample of Iodine-131 used at KNH has a half-life of 8 days. Using $N = N_0 \times (1/2)^n$, what fraction of the nuclei remains after 24 days?

Write your answer here:

Part 5: Closing the case: energy, uses and safety

(a) Explain where the energy released from the pitchblende ultimately comes from, using the idea of mass defect. (b) Describe the difference between nuclear fission and nuclear fusion. (c) Give two Kenyan applications of radioactivity from our lessons and explain how each uses a property of radiation or half-life. (d) Explain the ALARA principle and describe three ways to reduce exposure to radiation. (e) Write a short explanation, for a Year 9 student, of why some rocks are radioactive and why it matters for Kenya.

Write your answer here:

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining why the nucleus is unstable	Accurately explains nucleons, isotopes and the belt of stability, and clearly links the neutron-to-proton ratio of uranium-238 to spontaneous decay as a nuclear, not chemical, process.	Explains isotopes and instability correctly but the link to the belt of stability or to the nuclear nature of decay is incomplete.	Describes radioactivity without explaining nuclear instability or confuses isotopes with other ideas.
Radiation types and nuclear equations	Correctly compares alpha, beta and gamma, writes balanced equations with conserved A and Z, and explains decay arrows on the N vs Z plot.	Compares radiation types correctly and writes mostly correct equations with minor errors in balancing or in explaining movement on the plot.	Lists some properties but equations are unbalanced or missing, and no link to stability is made.
Detection, background radiation and half-life	Explains the GM tube process, background correction and half-life using the dice evidence, and correctly calculates the fraction remaining.	Explains most of detection and half-life correctly; the calculation or the use of simulation evidence has small gaps.	Gives vague statements about detection or half-life; calculation is missing or incorrect.

Energy source, applications and safety	Links mass defect and $E = mc^2$ to the rock's energy, distinguishes fission from fusion, and gives Kenyan applications and ALARA measures with correct reasoning.	Gives correct applications and safety measures, with only partial explanation of mass defect or of fission versus fusion.	Names applications or safety ideas without explanation, or with errors.
Communication and connection to the phenomenon	Writes clearly in logical order, uses scientific vocabulary precisely, and consistently ties each idea back to the pitchblende rock and Kenyan contexts.	Writing is mostly clear and uses key terms; links to the phenomenon are present but not consistent.	Writing is hard to follow, with little scientific vocabulary or connection to the rock.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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